EXCAVATOR TRENCHING PRODUCTION

When an excavator is used for trenching applications, a meaningful expression of work produced is the machine's trenching rate expressed in meters or lineal feet per hour or per day. Trenching rate depends on the earthmoving production of the excavator being used and the size of the trench being excavated. Earthmoving production converts to trenching production as follows:

$$\text{Lineal Meters of Trench per Hour} = \frac{\text{Cubic Meters Excavated per Hour}}{\text{Cubic Meters per Lineal Meter of Trench}}$$

$$\text{Lineal Meters of Trench per day} = (\text{Lineal Meters per Hour}) \times (\text{Trenching Hours per Day})$$

$$\text{Lineal Feet of Trench per Hour} = \frac{\text{Yd}^3 \text{ Excavated Per Hour}}{\text{Yd}^3 \text{ Per Lineal Foot of Trench}}$$

$$\text{Lineal Feet of Trench Per Day} = (\text{Lineal Ft Per Hour}) \times (\text{Trenching Hours Per Day})$$

For machines that work in trenching applications where they dig all of the time, the *Trenching Conversion Chart* provides easy conversion from m^3 (yd^3) per hour to m (lineal feet) per hour, if the excavating rate m^3/hr (yd^3/hr) and trench volume m^3/m (yd^3/ft) are known. The following examples demonstrate how the Trenching Conversion Chart can be used.

Example problem (Metric)

Contractor estimates that a Track 30T Excavator will produce 200 Lm^3/hour. Trench survey shows that the trench contains 2.5 Lm^3/meter. What trenching rate will the Track 30T produce?

Solution: Enter the horizontal axis of the Trenching Conversion Chart at 200 m^3/Hour and move up to the 2.5 m^3/m diagonal line. Then move left to the vertical axis of chart and read answer of 80 m/hour.

● ● ●

Example problem 2 (Metric)

Contractor knows he must produce 1000 meters of trench in every 10 hour work day. Survey shows that trench contains 1.5 Bm^3 per lineal meter and soil swell factor is estimated at 30%. How much earthmoving production will the excavator have to provide in order to get the job done on time assuming a 50 min work hour? What Cat excavator will provide needed production at 6 meter maximum depth in sandy loam soil?

Solution: Determine trenching requirement 1000 meters in 10 hrs = 100 m/h. Convert Bm^3 to Lm^3 (excavator handles Lm^3) 1.5 Bm^3/m × 1.30 = 2.0 Lm^3/m. Enter vertical axis of trenching conversion chart at m/h and travel horizontally to diagonal line representing 2.0 m^3/m. Next move down to horizontal axis and read answer to 200 Lm^3/50 min hr. Convert 200 Lm^3/ 50 min hr to Lm^3/ 60 min hr = 200 = 241 Lm^3/60 min hr.

Production estimating tables in this section show that 241 Lm^3/60 min hr is within the capability of a Track 30T Excavator. Job should then be checked for reach and lifting requirements to make sure that a Track 30T could handle these aspects of the work.

● ● ●

Example problem (English)

Contractor estimates that a Track 30T Excavator will produce 250 LCY/Hour. Trench survey shows that the trench contains 2.5 LCY/Foot. What trenching rate will a Track 30T produce?

Solution: Enter the horizontal axis of the Trenching Conversion Chart at 250 yd^3/hr. Then move to the vertical axis of chart and read answer of 100 ft/hr.

The Trenching Conversion Chart can also be used to determine the required excavating rate if the contractor can define his trenching production requirement and the trench volume per lineal foot.

● ● ●

Example problem 2 (English)

Contractor knows he must produce 1000 ft of trench in every 10 hr work day. Survey shows that trench contains 1.6 BCY per lineal ft and soil swell factor is estimated at 25%. How much earthmoving production will excavator have to provide in order to get the job done on time assuming 50 min work hour? What Cat model will provide needed production at 8 ft depth in sandy loam soil?

Solution: Determine trenching requirement — 1000 ft in 10 Hrs. = 100 ft/hr
Convert BCY to LCY — 1.6 BCY/ft × 1.25 = 2.0 LCY/ft

Enter vertical axis of trenching conversion chart at 100 ft/hr and travel over to diagonal line representing 2.0 yd³/ft. Next move down to horizontal axis and read answer of 200 LCY/50 min hr.

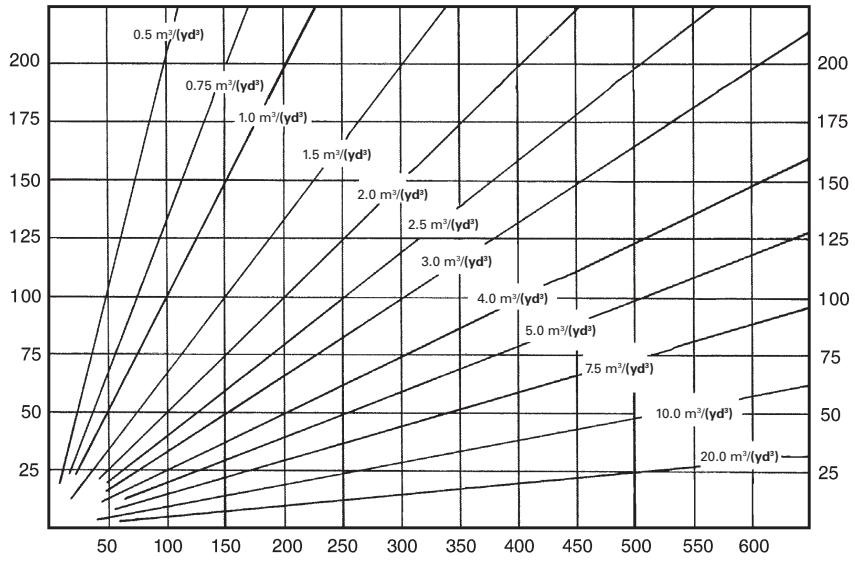
Convert 200 LCY/50 min hr to LCY/60 min hr =

$$\frac{200}{0.83} = 241 \text{ LCY/60 min hr}$$

Production estimating tables in this section show that 241 LCY/60 min. hr. is within capability of a 30T Excavator. Job should then be checked for reach and lifting requirements to make sure that the 30T could handle these aspects of the work.

● ● ●

TRENCHING CONVERSION CHART — CUBIC METERS (yd³) PER HOUR TO METER (ft) PER HOUR



m (ft.)= m³/(yd³) Hr.
Hr. = m³/(yd³) m (ft.)

Values in m³/m or yd³/ft

If excavating rate has been calculated in Bm³/h use Bm³/m for Trench Volume/m.
" " " " " " " Lm³/h use Lm³/m for Trench Volume/m.
" " " " " " " BCY/Hr use BCY/ft for Trench Volume/ft.
" " " " " " " LCY/Hr use LCY/ft for Trench Volume/ft.

Estimating Bucket Size

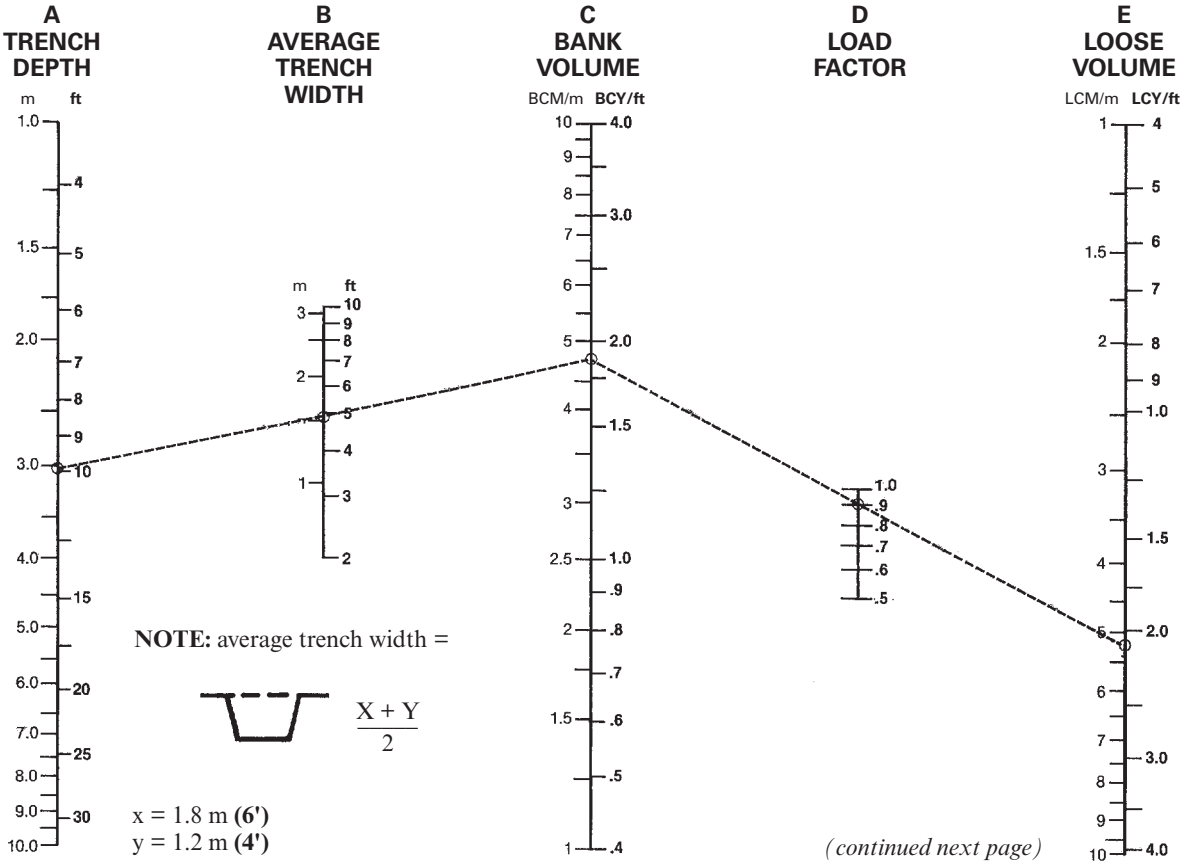
In addition to the trenching calculations on the previous pages, an alternative method of figuring trenching production is the nomograph. Shown on the following pages, this particular nomograph can be used for estimating bucket size when given trench dimensions and linear production rate. The nomograph is quicker and easier than the preceding example because it does not require as many calculations, yet the accuracy is about the same within the normal limits of input data.

Be careful when entering and reading data from the nomographs because some scales increase from bottom to top, while others are the reverse. Do not be overly concerned with the precision as affected by pencil line width or reading to the hundredth of a m^3 (yd^3). Remember that bucket fill factor, material density and cycle time are at best close estimates.

Example problem:

A sewer contractor owns a 30T with 2 piece boom and short stick. He wants to bid a contract for a 3.1 m (10') deep trench which measures 1.8 m (6') at the top and 1.2 m (4') at the bottom. He must dig 9 m/hr (30 ft/hr) to finish on time. The material is sand and gravel with a load factor of 0.90 and 100% bucket fill factor. He works 54 minutes per hour, half the time digging and half setting pipe. Cycle time is estimated at 23 seconds which includes a 90° swing angle.

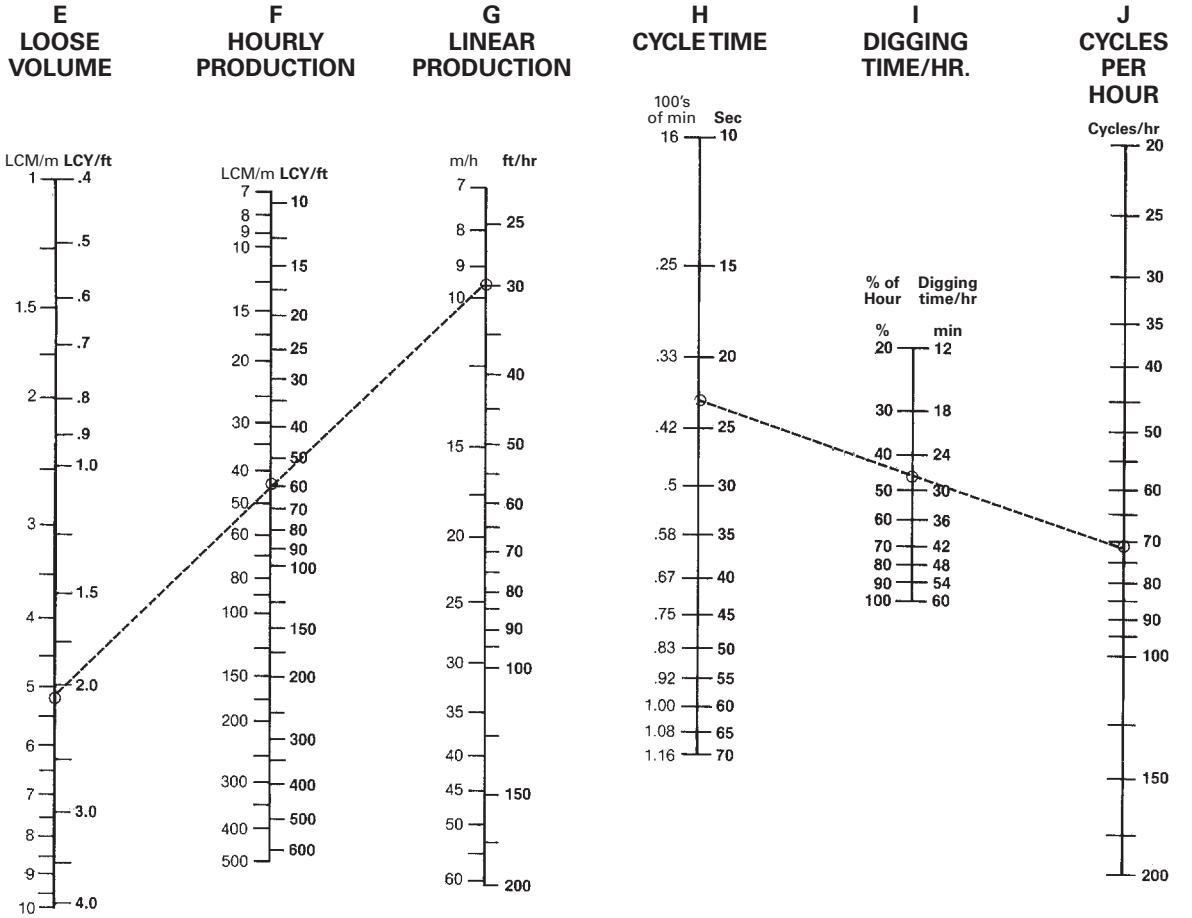
1) Enter trench depth 3.1 m (10') on scale A and average trench width 1.5 m (5') on scale B.
2) Connect A and B and extend to scale C for bank volume per m (ft).
3) Enter estimated load factor (0.90) on scale D.
4) Connect C & D and extend to scale E for loose volume per m (ft).



(continued next page)

(Get loose volume from scale E and enter on this page scale E.)

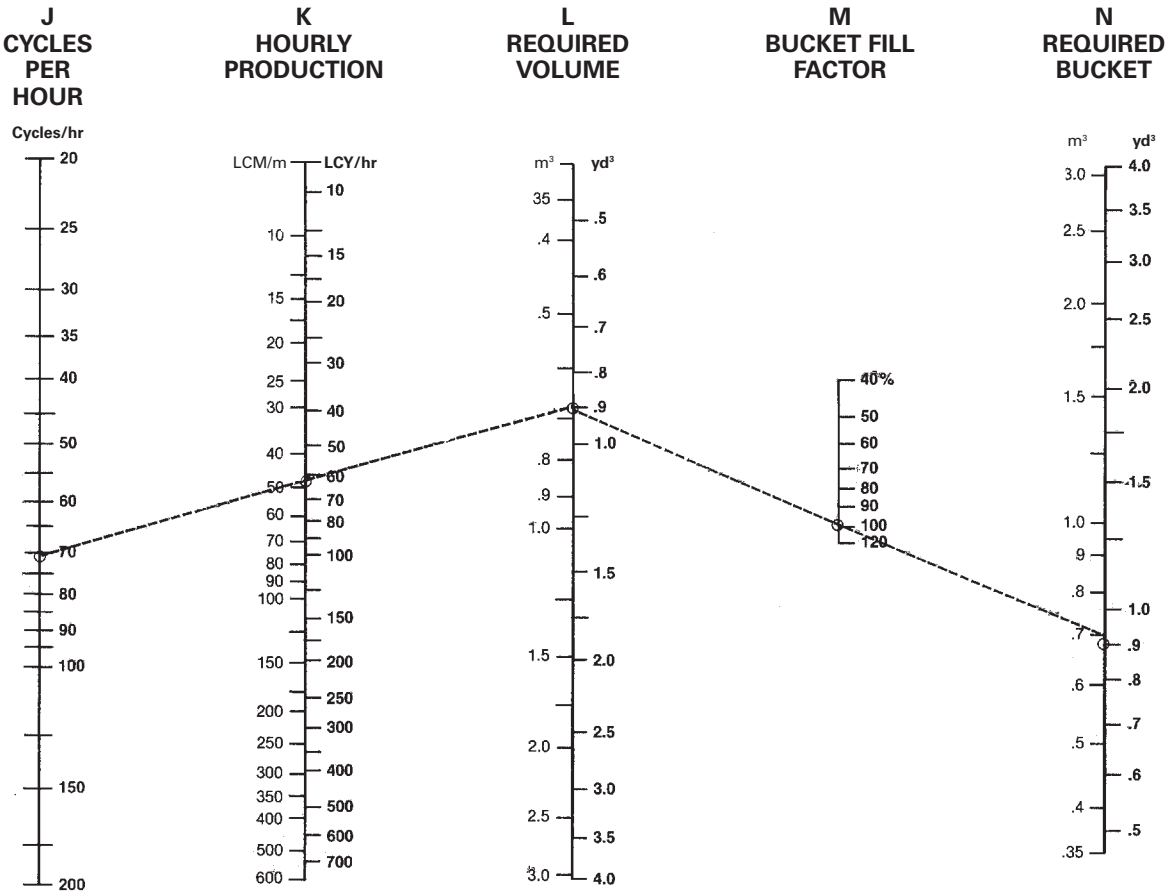
5) Enter required linear production rate 9 m/h (30 t/hr) on scale G.
6) Connect E and G. Transfer hourly production rate from scale F to scale K (next page).
7) Estimate cycle time (23 sec) based on anticipated conditions and enter on scale H.
8) Estimate hourly digging time (27 min) and enter on scale I.
9) Connect H through I to scale J for cycles per hour.

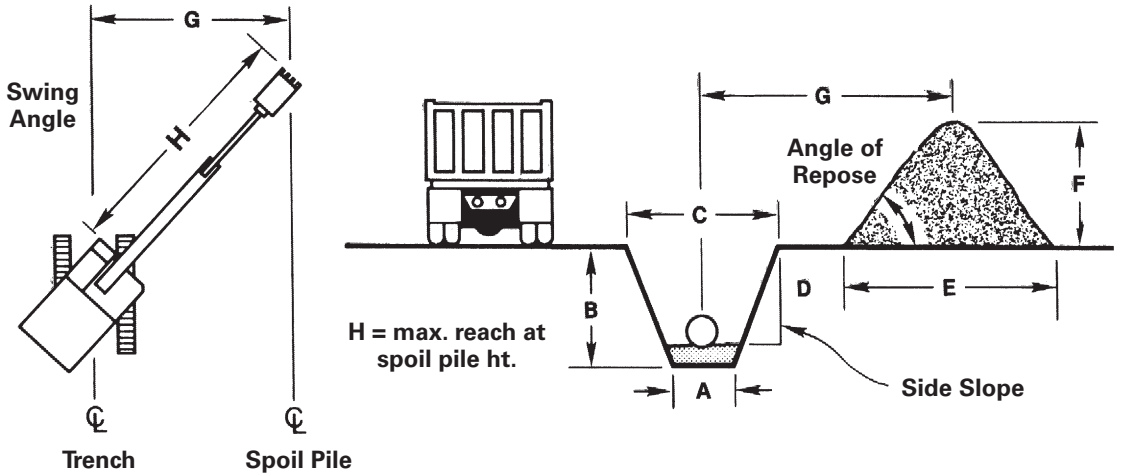


(Get cycles per hour from scale J and enter on this page scale J.)

10) Connect J through K to scale L for required volume per cycle.
11) Enter estimated bucket fill factor (100%) on scale M.
12) Connect L through M to scale N for required bucket size.

NOTE: Ensure bucket width does not exceed minimum trench width and also that weight of bucket and payload does not exceed machine working weight capacity (see lift capacity charts in this section).





Excavation Volumes Per Meter or Foot of Trench Length

Metric version

Bank m^3/meter = (Trench end area m^2) × (one m)
Trench volume (Bm^3/m) = $^1/_2$ (A + C) × B
Spoil pile volume (Lm^3/m) = (Bm^3/m) × (1.00 + % Swell)

English version

$$\text{Bank } yd^3/\text{foot} = \frac{(\text{Trench end area } ft^2) \times (\text{one ft})}{27}$$

$$\text{Trench volume (BCY/ft)} = \frac{^1/_2\,(A + C) \times B}{27}$$

Spoil pile volume (LCY/ft) = (BCY/ft) × (1.00 + % Swell)

The following table provides a general guide to trench bottom width for various outside diameters of pipe.

Pipe Diameter		Trench Width		Pipe Diameter		Trench Width	
mm	ft/in	m	ft/in	mm	ft/in	m	ft/in
102	**4"**	0.49	**1'7"**	1524	**5'0"**	2.59	**8'6"**
152	**6"**	0.55	**1'10"**	1676	**5'6"**	2.80	**9'2"**
203	**8"**	0.61	**2'0"**	1829	**6'0"**	3.05	**10'0"**
254	**10"**	0.70	**2'4"**	1981	**6'6"**	3.26	**10'8"**
305	**12"**	0.76	**2'6"**	2134	**7'0"**	3.47	**11'5"**
381	**15"**	0.91	**3'0"**	2286	**7'6"**	3.69	**12'1"**
457	**18"**	1.03	**3'5"**	2438	**8'0"**	3.93	**12'11"**
533	**1'9"**	1.16	**3'10"**	2591	**8'6"**	4.15	**13'7"**
610	**2'0"**	1.25	**4'1"**	2743	**9'0"**	4.36	**14'4"**
686	**2'3"**	1.37	**4'6"**	2896	**9'6"**	4.54	**14'11"**
838	**2'9"**	1.58	**5'2"**	3048	**10'0"**	4.75	**15'7"**
914	**3'0"**	1.70	**5'7"**	3200	**10'6"**	4.99	**16'5"**
1067	**3'6"**	1.92	**6'4"**	3353	**11'0"**	5.21	**17'1"**
1219	**4'0"**	2.13	**7'0"**	3505	**11'6"**	5.43	**17'10"**
1372	**4'6"**	2.38	**7'10"**	3658	**16'2"**	5.64	**18'6"**

NOTE: Trench widths based on 1.25 Bc + 1.0 where Bc is the outside diameter of the pipe in feet.
Table courtesy of American Concrete Pipe Association

Trenching Production with Pipesetting

On many sewer construction jobs the excavator does more than just dig the trench. Other tasks include handling the shoring system, placing bedding material, and lowering the pipe. The normal work procedure is to open a section of trench and then stop and make a pipe installation before going on to dig the next section of trench. At that point the key to trenching production is the total amount of time required to install each section of pipe. Pipe installation time can be broken down as follows:

Digging time + other time = Total pipe installation time

Total Pipe Installation Time	Pipe Installed Per Hour
60 min	1 Pipe/hr
30 min	2 Pipe/hr
15 min	4 Pipe/hr
10 min	6 Pipe/hr

Digging Time can be calculated once the trenching rate has been calculated using the methods described earlier in this section. Once Digging Time has been calculated, it can be added to an estimate of "Other Time" to determine Total Pipe Installation Time. "Other Time" can be estimated based on a contractor's judgment, experience, or actual measurement on a job. The following formula and table relate the trenching rate of the excavator to the time required to open a section of trench for pipe of various lengths.

$$\text{Digging Time (Min.)} = \frac{\text{Pipe Length (ft)}}{\text{Trenching Rate (ft/hr)}} \times 60 \text{ (Min/hr)}$$

Trenching Rate Ft. Per Hour	Time Required to Dig for Pipe of Various Lengths							
	8 ft Pipe		12 ft Pipe		16 ft Pipe		20 ft Pipe	
	Hours	Min.	Hours	Min.	Hours	Min.	Hours	Min.
20 ft/hr	0.400	24.00	0.600	36.00	0.800	48.00	1.000	60.00
40 ft/hr	0.200	12.00	0.300	18.00	0.400	24.00	0.500	30.00
60 ft/hr	0.130	8.00	0.200	12.00	0.260	16.00	0.333	20.00
80 ft/hr	0.100	6.00	0.150	9.00	0.200	12.00	0.250	15.00
100 ft/hr	0.080	4.80	0.120	7.20	0.160	9.60	0.200	12.00
120 ft/hr	0.060	4.00	0.100	6.00	0.120	7.20	0.167	10.00
140 ft/hr	0.057	3.43	0.086	5.14	0.114	6.86	0.143	8.57
160 ft/hr	0.050	3.00	0.075	4.50	0.100	6.00	0.125	7.50
180 ft/hr	0.044	2.66	0.067	4.00	0.089	5.33	0.111	6.67
200 ft/hr	0.040	2.40	0.060	3.60	0.080	4.80	0.100	6.00

This table can be used to show how an excavator that is capable of more trenching production will provide significant advantages even on jobs where the machine does not dig all of the time. Consider 12,000' job with 12' sections of pipe (1000 pipe to be installed). Excavator "A" can work at 60 ft/hr while Excavator "B" is capable of producing 120 ft/hr. Table shows that Excavator "B" will only take 0.10 hr to do the same work. This means that over the course of installing the 1000 pipe the more productive machine will save 0.10 hr/pipe or 100 hours of working time.

Example problem (English)

The following example shows how trenching production can be calculated on a job where the excavator is also required to set pipe. This example is based on the assumption that the excavator's earthmoving rate and the pipe installation time have already been estimated by the contractor.

Problem: Contractor estimates that the 50T Excavator will be able to produce 500 LCY/60 min. hr. Survey shows that an average cross section trench contains 3.2 BCY/ft and swell factor for sandy clay soil is estimated at 25%. How much trenching production can a contractor expect; assuming it takes 10.0 min. to install each 20 ft length of pipe after trench has been opened. Also assume 83% job efficiency — 50 min. work hour and 8 work hours out of a 9 hour shift. (0.5 hours for lunch and two 15 minute breaks.)

Solution:

Convert trench volume to LCY/ft:

$$1.25\ (3.2\ \text{BCY/ft}) = 4.0\ \text{LCY/ft}$$

Convert Earthmoving rate to Trenching rate:

$$\frac{500\ \text{LCY/hr}}{4.0\ \text{LCY/ft}} = 125\ \text{ft/hr}$$

Calculate digging time for each pipe:

$$\frac{20\ \text{ft/pipe}}{125\ \text{ft/hr}} = 0.16\ \text{hr/pipe} = 9.6\ \text{min}$$

Calculate pipe installation time:

Digging time	=	9.6 min
Other time	=	10.0 min
Pipe Installation time	=	19.6 min

Calculate pipe installations/hour:

$$\frac{60\ \text{min/hr}}{19.6\ \text{min/pipe}} = 3.06\ \text{pipe/hr}$$

Calculate max. pipe installations/day:

$$8\ \text{hrs}\ (3.06\ \text{pipe/hr}) = 24.48\ \text{pipe/day}$$

Actual pipe/day:

$$0.83\ (24.48\ \text{pipe/day}) = 20.3 - 20\ \text{pipe/day}$$

Actual feet/day:

$$(20\ \text{pipe/day}) \times (20\ \text{ft/pipe}) = 400\ \text{ft/day}$$

HYDROMECHANICAL ATTACHMENTS

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Lifecycle Value

Performance Line



- Oil Fired Operating Cycle
- Silenced as Standard
- World class production
- Automatic Shut Off (blank fire protection)
- Slip – fit/Rotatable lower tool bushing
- Machine protection: buffer system
- Machine protection: accumulator
- Optional machine or hammer mount auto – lube system
- Optional wear package for housing

GC Silenced



- Gas Fired Operating Cycle
- Silenced as Standard
- Industry standard production
- Automatic Shut Off (Blank fire protection)
- Slip-fit lower tool bushing
- Machine protection: buffer system
- Machine protection: accumulator
- Optional machine or hammer mount auto – lube system
- Manual adjustable piston stroke selector

GC Line



- Gas Fired Operating Cycle
- Industry standard production
- Slip-fit lower tool bushing
- Machine protection: accumulator
- Manual adjustable piston stroke selector

Tool Family	Standard Tools		
Tool Type	Chisel	Moil	Blunt
Road building/construction			
Breaking of road surface	○	○	
Breaking uneven bedrock to lay a road	●	●	
Primary breaking to prepare road bed	○	○	
Trench excavation for drainage	○	○	
Demolition of bridges	○	○	○
Heavily reinforced bridge pillars	○	○	●
Making holes (for traffic signs, lamp posts)		○	
Breaking of frozen ground	●	●	
Demolition/housing development			
Demolition of concrete walls, roofs, floors	●	●	●
Demolition of light, reinforced concrete (<20")	○	○	○
Brick walls	○	○	○
Rock trenches for mains/water supply/utilities	○	○	
Rock excavation for foundation	○	○	
Mass excavation of rock for industrial building bases	○	●	
Massive reinforced concrete foundations	○	●	
Separating rebar from concrete (for recycling)	○	○	●
Quarrying/open cast mining			
Secondary boulder breaking			○
Primary breaking of rock	●	●	
Breaking over sizes on a crusher/feeder/feed chute	○	○	○
Underground applications			
Scaling			
Metallurgical applications			
Breaking of slag in casting ladles			
Breaking of slag in converter openings	●	●	
Cleaning of castings			
Breaking of massive steel slug			
Breaking of aluminum electrolyze slug	●	●	
Other applications			
Demolition/Rock breaking under water	●	●	

● Optimal ○ Acceptable Not Recommended

H110 S-H180 S Hammer Features:

- **Unique Suspension System** — Improved recoil, support and guidance protects the carrier, increases hammer durability. Entire power cell is secured firmly inside housing. Noise suppression, operator feel and control is improved.
- **Auto Shut Off (ASO)** — Instantly stops the piston when breaking through material. Prevents blank firing, which is a top cause of hammer wear. Reducing wear improves maintenance and more productive hours of work.
- **Accumulator** — Self-contained membrane accumulator designed for long life. Port is accessible while hammer is mounted on the machine making testing and recharging a routine task achievable in the field.
- **Hydraulic Valves** — A Pressure Control Valve (PCV) maintains maximum hydraulic pressure to ensure the hammer delivers all blows at full power. PCV can be easily checked and adjusted from outside the hammer in about 30 minutes. A check valve (not shown) isolates harmful pulsation spikes from the carrier hydraulic circuit.
- **Autolube Connection and Grease Channel** — Provides grease to the upper and lower tool bushings to ensure proper greasing, longer life for bushings and tool.
- **Seal Carrier** — Contains special high performance seals to extend leak-proof operation.
- **Piston** — Long piston transfers a long shock wave into the rock. Tool-piston diameters are matched for maximum energy transfer.
- **Tie-Rods** — Larger threads improve load carrying capability, durability and reliability.
- **Cylinder** — Engineered to be durable and reliable with minimal maintenance and downtime.
- **Upper Tool Bushing** — Guides the tool to optimize in-line piston to tool contact.
- **Tool Retaining Pins and Keepers** — Tool removal process is simplified, achievable with common hand tools. Removal time reduced by 40% over previous models.
- **Lower Tool Bushing** — As bushing reaches the wear limit, it can be easily rotated (90°) or replaced to bring it back into specification. Dust seals keep contaminants out.

NOTE: Internal components of hammers are machined to close tolerances and require clean oil with full lubricating properties. When operating in high ambient temperatures or extreme temperature applications (e.g. foundries), higher viscosities are recommended to extend hammer life and improve performance. Hammers tend to shear multigrade mineral oil so that oil viscosity decreases. Contamination, water in oil, and decreased viscosity lead to earlier oil deterioration and the need for more frequent oil changes than normally recommended for the excavator. Extra care should be taken to avoid the entry of dust or dirt when installing or removing a hammer in the field.

Large Hammer Tools

Chisel	Hard Rock Chisel	Moil	Blunt	Super Blunt
C	C	M	B	B
				

	H110 S	H115 S	H120 S	H130 S	H140 S	H160 S	H180 S
Road building/construction							
Breaking of road surface	C, M	C, M	C, M	C, M	C, M	C, M	C, M
Breaking uneven bedrock to lay a road	C, M	C, M	C, M	C, M	C, M	C, M	C, M
Primary breaking to prepare road bed					C, M	C, M	C, M
Trench excavation for drainage	C, M	C, M	C, M	C, M	C, M	C, M	C, M
Demolition of bridges	C, M	B, C, M	B, C, M	B, C, M	B, C, M	B, C, M	B, C, M
Heavily reinforced bridge pillars					B	B	B
Making holes (for traffic signs, lamp posts)	M	M	M	M	M	M	M
Breaking of frozen ground	C, M	C, M	C, M	C, M	C, M	C, M	C, M
Demolition/housing development							
Demolition of concrete walls, roofs, floors	C, M	B, C, M	B, C, M	B, C, M	B, C, M	B, C, M	B, C, M
Demolition of light, reinforced concrete [<0.5 m **(<20")**]	M	B, M	B, M	B, M			
Brick walls	C, M	B, C, M	B, C, M	B, C, M	B, C, M	B, C, M	B, C, M
Rock trenches for mains/water supply/utilities	C, M	C, M	C, M	C, M	C, M	C, M	C, M
Rock excavation for foundation	C, M	C, M	C, M	C, M	C, M	C, M	C, M
Mass excavation of rock for industrial building bases				C, M	C, M	C, M	C, M
Massive reinforced concrete foundations					M	M	M
Separating rebar from concrete (for recycling)	C, M	B, C, M	B, C, M	B, C, M	B, C, M	B, C, M	B, C, M
Quarrying/open cast mining							
Secondary boulder breaking	B	B	B	B	B	B	B
Primary breaking of rock				C, M	C, M	C, M	C, M
Breaking over sizes on a crusher/feeder/feed chute	C, M	B, C, M,	B, C, M	B, C, M	B, C, M	B, C, M	
Underground applications							
Scaling	C	C					
Metallurgical applications							
Breaking of slag in casting ladles	C, M	C, M					
Breaking of slag in converter openings	C, M	C, M	C, M	C, M	C, M		
Cleaning of castings	C, M	C, M					
Breaking of massive steel slag						C, M	C, M
Breaking of aluminum electrolyze slag	C, M	C, M	C, M	C, M	C, M		
Other applications							
Demolition/rock breaking under water		C, M	C, M	C, M	C, M	C, M	C, M

	H95 S	H110 S	H115 S	H120 S	H130 S	H140 S	H160 S	H180 S
Excavators								
307D	•							
308D CR/308E CR SB/308E2 CR SB	•							
311D RR	•	•						
312D/312D L/312E/312E L	•	•	•					
314D CR/314E CR		•	•					
315D		•	•	•				
316E		•	•	•				
318E			•	•	•			
319D			•	•	•			
320D/320E			•	•	•			
320D RR/320E RR			•	•	•			
321D LCR				•	•			
324D				•	•	•		
324E				•	•	•		
328D LCR					•	•		
329D/329E				•	•	•		
336D/336E/336EH						•	•	
345D							•	•
349E							•	•
365C								•
374D								•

*Installation of add-on optional counterweight to machine is required.
**Hydraulic flows and pressures must be checked to verify they match the requirements for the hammer being installed.
Note: Caterpillar recommends the use of a suitable shield/guard system to insure operator has adequate protection from flying debris.
Note: These matches are for general reference purposes for Cat machines only. When special boom and quick coupler arrangements are used, these matches may not apply.
Note: When matching hammers to competitive carriers, selection should be made by carrier weight. Refer to the carrier range at the top of the table in order to determine the correct match.

Hammer Model	H110 S	H115 S	H120 S	H130 S
Machines (WHEX)				
M314	■	■		
M315	■	■		
M316	■	■	■	
M317	■	■	■	
M318	■	■	■	
M319	■	■	■	
M320		■	■	■
M322		■	■	■

Hammer Model	H110 S	H115 S	H120 S	H130 S	H140 S	H160 S	H180 S	H190 S	H215 S
Machines (THEX)									
312 GC	■	■							
313 GC	■	■							
313	■	■							
315 GC	■	■							
315	■	■							
316 GC	■	■							
317 GC	■	■	■						
317	■	■	■						
320 GX		■	■	■					
320 GC		■	■	■					
320		■	■	■					
323 GX			■	■					
323 GC			■	■					
323		■	■	■					
325		■	■	■					
325 Tunneling			■	■	■				
326 GC			■	■	■				
326			■	■	■				
330 GC			■	■	■				
330			■	■	■				
335			■	■	■				
336 GC					■	■			
336					■	■			
340					■	■			
340 UHD					■	■			
345 GC						■	■	■	
349						■	■	■	
352						■	■	■	
352 UHD						■	■	■	
374							■	■	■
395									■

Please refer to Caterpillar Reference Guide GEJQ9332-02. Some model generations may NOT be official configuration.

Note 1: Caterpillar recommends the use of a suitable shield/guard system to ensure operator has adequate protection from flying debris.

Note 2: These matches are for general reference purposes for Cat machines only. When special boom and quick coupler arrangements are used, these matches may not apply.

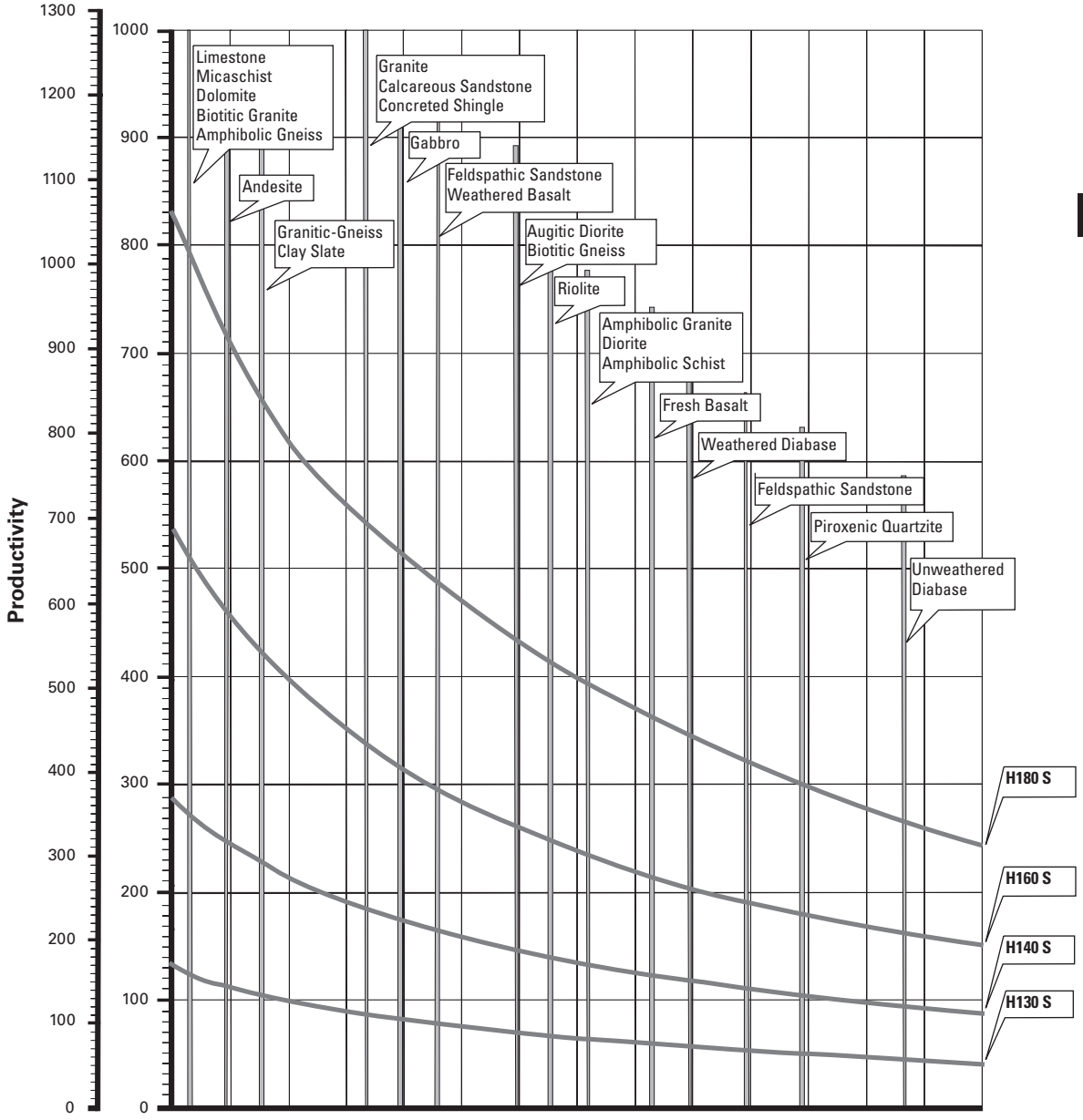
Note 3: When matching hammers to competitive carriers, selection should be made by career weight. Refer to the carrier range at the top of the table in order to determine the correct match.

Productivity Rates: 8 hour shift

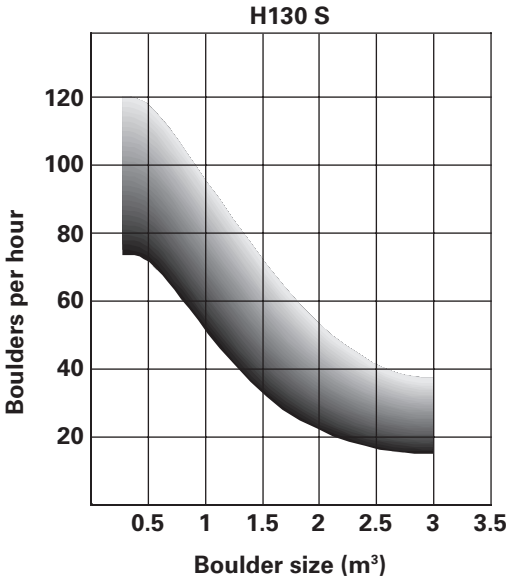
Production rates listed are for general estimation purposes only and must not be used to guarantee any production figure to the customer. The actual working results may vary according to the quality and structure of the material to be broken, required degree of material size reduction, installation, condition of the carrier, conditions at the work site, haulage of the broken material, skills of the operator, etc.

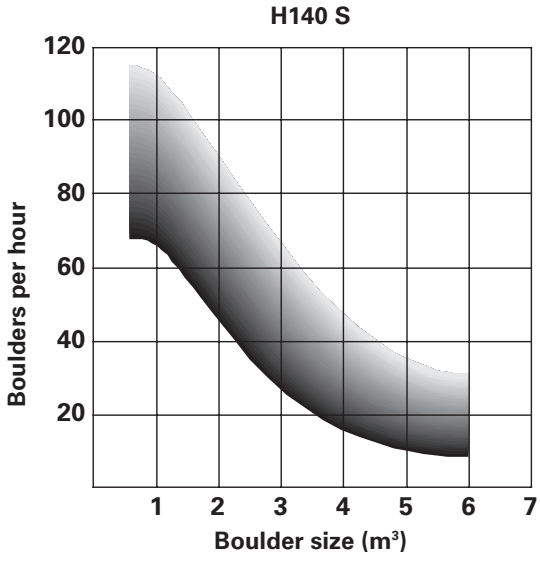
Hammer Models	Non-Reinforced Concrete		Reinforced Concrete		Sedimentary Rock		Volcanic Rock	
H110 S	99-214 m^3	**130-280 yd^3**	96-134 m^3	**125-175 yd^3**	84-121 m^3	**110-250 yd^3**	42-99 m^3	**55-130 yd^3**
H115 S	115-287 m^3	**150-375 yd^3**	107-184 m^3	**140-240 yd^3**	126-229 m^3	**165-300 yd^3**	57-115 m^3	**75-150 yd^3**
H120 S	153-344 m^3	**200-450 yd^3**	122-229 m^3	**160-300 yd^3**	153-260 m^3	**200-340 yd^3**	84-153 m^3	**110-200 yd^3**
H130 S	210-375 m^3	**275-490 yd^3**	153-268 m^3	**200-350 yd^3**	191-306 m^3	**250-400 yd^3**	103-210 m^3	**135-210 yd^3**
H140 S			191-497 m^3	**250-650 yd^3**	229-535 m^3	**300-700 yd^3**	115-268 m^3	**150-350 yd^3**
H160 S			229-650 m^3	**300-850 yd^3**	268-688 m^3	**350-900 yd^3**	153-459 m^3	**200-600 yd^3**
H180 S			295-1301 m^3	**385-1705 yd^3**	337-1345 m^3	**440-1760 yd^3**	210-757 m^3	**275-990 yd^3**

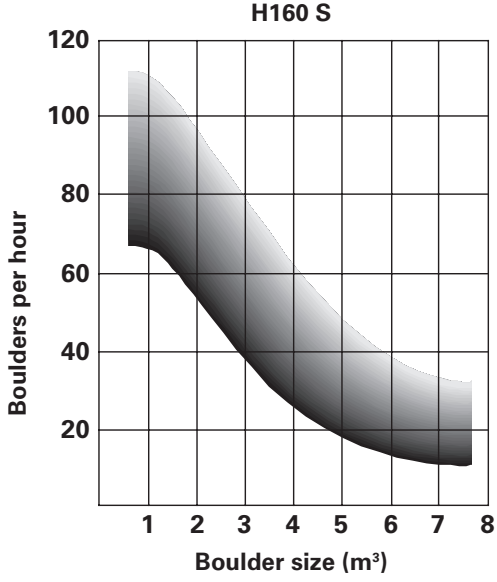
The figures are for comparison and evaluation purposes only. Results will vary depending on operator, carrier and job conditions.

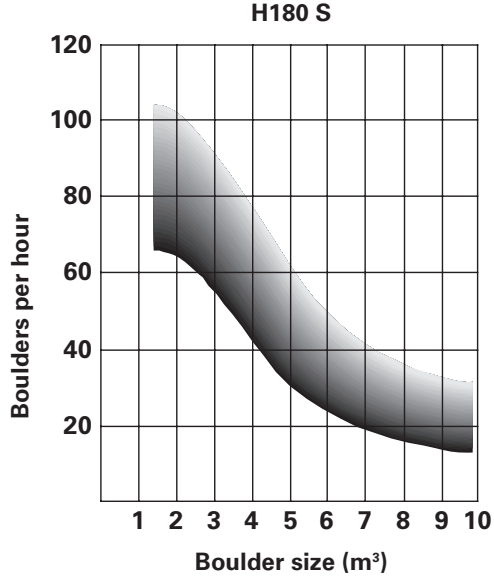


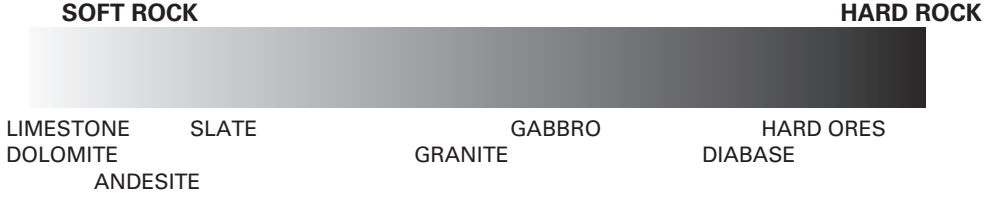
Bedding thickness 100-200 cm (40-80") or closely spaced vertical fractures


H130 S
Boulders per hour
120
100
80
60
40
20
0.5
1
1.5
2
2.5
3
3.5
Boulder size (m³)


H140 S
Boulders per hour
120
100
80
60
40
20
1
2
3
4
5
6
7
Boulder size (m³)


H160 S
Boulders per hour
120
100
80
60
40
20
1
2
3
4
5
6
7
8
Boulder size (m³)


H180 S
Boulders per hour
120
100
80
60
40
20
1
2
3
4
5
6
7
8
9
10
Boulder size (m³)


SOFT ROCK
HARD ROCK
LIMESTONE
SLATE
GABBRO
HARD ORES
DOLOMITE
GRANITE
DIABASE
ANDESITE

- **Piston Stroke Adjuster** — Manual adjustment located on the side of the hammer power cell will increase piston speed and lower the power of each piston blow. This feature allows the owner/operator to match the speed and power of hammer for a particular application. Less frequency, more power for tougher jobs, increased frequency, less power for lighter jobs.
- **Automatic Shut Off (ASO)** — This feature offers blank firing protection by stopping the piston from cycling internally when there is an absence of material to be broken under the tool. Eliminating blank firing protects the hammer by reducing internal stresses and heat, which lowers overall hammer owning and operating costs. Additionally, ASO will protect rental fleet hammers where operator skill level and expertise may vary.
- **Simplistic Design/Gas Fired Hammer** — The design of the Cat GC hammer line is well proven with our customers. The simplistic design allows shorter rebuild time and lower owning and operating costs. The Gas fired operating cycle, delivers expected industry production and power that customers have come to expect from Cat Hammers.
- **Hammer Auto-Lube Options** — The most critical maintenance requirement for hammers is that it receives Cat Hammer Paste. In addition to manually greasing, customers and dealers have the option of installing either a carrier mounted auto-lube system or a hammer mounted auto-lube system.
- **Slip Fit Lower Tool Bushing** — The lower tool bushing is a “slip-fit” design which allows for field replacement. The lower tool bushing is critical for piston/tool alignment and ultimately greater uptime with the GC hammer. The design reduces overall service time and lowers owning and operating costs.

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Machines (WHEX)	M313	M314	M315	M316	M317	M318	M320	M322
Hammer Model								
H115GC S/H115GC								
H120GC S/H120GC								

Machines (THEX)	311	312	313	314	315	316	318	320	323	325	326	329-330	335	336-340
Hammer Model														
H110GC S/H110GC														
H115GC S/H115GC														
H120GC S/H120GC														
H130GC S/H130GC														
H140GC S/H140GC														

Please refer to Caterpillar Reference Guide GEJQ9332-02. Some model generations may NOT be official configuration.

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Hammer Models	Non-Reinforced Concrete		Reinforced Concrete		Sedimentary Rock		Volcanic Rock	
H110GC S	87-190 m^3	**115-250 yd^3**	85-119 m^3	**111-157 yd^3**	74-107 m^3	**97-122 yd^3**	35-87 m^3	**47-116 yd^3**
H115GC S	102-257 m^3	**132-335 yd^3**	97-164 m^3	**125-215 yd^3**	111-205 m^3	**148-265 yd^3**	50-101 m^3	**66-132 yd^3**
H120GC S	137-308 m^3	**177-402 yd^3**	107-205 m^3	**143-265 yd^3**	136-231 m^3	**157-305 yd^3**	75-135 m^3	**97-157 yd^3**
H130GC S	188-335 m^3	**245-435 yd^3**	137-238 m^3	**176-312 yd^3**	173-279 m^3	**222-357 yd^3**	90-188 m^3	**121-189 yd^3**
H140GC S			168-445 m^3	**222-589 yd^3**	201-482 m^3	**267-625 yd^3**	100-238 m^3	**135-312 yd^3**

Principles of Selection

Key to the successful sale of a hammer is proper hammer selection.

Background Information

Collection of background information is the first step. The following information will assist in being sure the customer receives the correct hammer and that he has a positive hammer experience. The following issues should be examined…

1. If any, what brand and model hammer was previously used and how did the hammer perform?
2. What % of time will the hammer be used on the machine?
3. Will the hammer be used in primary breaking or secondary breaking? (mainly an issue for large hammers)
4. What machine will the hammer be used on and what are the hydraulic flow and pressures of this machine?
5. What is the type of material to be broken and production required from the hammer? (best to obtain this from the end user but a table is available at the end of this section)

Hammer Selection Process

1. Using Cat carrier matching guide, identify 2 or 3 possible hammers for your application (for competitive carriers use carrier weight class as reference).
2. Compare machine/carrier flow and pressures to those of the hammer candidates to validate compatibility. Eliminate hammers outside carrier specs.
3. If hammer is to be used in primary breaking consider larger of hammer candidates.
4. Check productivity guidance tables within this section. Identify hammer most compatible with requirements.
5. Determine if the application requires special hammer modifications, i.e. steel mill, underwater, tunneling, etc.

Other Issues

Once the hammer has been chosen, other elements need to be considered to have a successful hammer experience.

1. Select the correct hammer tool for the application (see tool application chart in this section).
2. Check to be sure the correct hammer bracket and hoses are specified. Be sure correct carrier oil is specified for hammer use (particularly important in high ambient areas).
3. Consider supplemental carrier cooling in areas of high ambient temperature.

Actual operating pressure and back pressure MUST be checked when the hammer is fitted to the carrier (just as important if the hammer goes on a competitive carrier or is installed by the contractor at his shop).

Guarding Recommendation

Hammers used in hazardous applications like demolition, quarrying, and scaling, can create a need for special operator guarding due to flying objects. When using a hammer, additional protective devices such as a front screen, Falling Object Guarding System (FOGS, includes top and front guarding), thick polycarbonate windshields or a combination of these is recommended by Caterpillar. Contact your Cat dealer for operator guarding options on your machine.

Mobile Scrap and Demolition Shears Features:

Cat scrap and demolition shears are designed to make you money. An innovative, robust design with fast cycle times provides the production and reliability you need to be profitable. Cat shears can handle everything from structural steel to mixed scrap. Cat shears are designed to get the job done.

Not all attachments are available in all regions. Consult your Cat dealer for specific configurations available in your region.

Features:

Cut More, Cut Faster

- Shears are designed as a system solution to cut more tons per day and make you more money.
- Dual apex jaw increases cut efficiency by 15 percent.
- Accurately place the jaws in optimum cutting position using the standard 360° rotator on the S3000 Series.
- Power is consistent through the entire cutting cycle.
- Increase cutting efficiency with tapered spacer plates which reduce jamming and drag.
- The cylinder rod is completely protected inside the frame reducing downtime and the risk of damage.
- The jaw relief area allows material to fall away freely without hindering the next cutting cycle.

Long-Lasting Quality

- Reduce maintenance costs with the new Bolt-on Piercing Tip. Dowels and bushings divert stress, keeping the tip tight and in one piece.
- Work confidently. Major hydraulic components are built with a 4 to 1 safety factor and can withstand pressure spikes up to 1378 bar (20000 psi).
- Increase overall durability with the solid plate construction of the upper jaw.
- The housing is designed with a 60 percent safety factor in yield strength.

Easy to Maintain

- Easily and safely complete maintenance tasks. Shears can be greased at ground level along with the machine stick cylinder pin and boom pin.
- Reduce downtime with field serviceable swivels. Removing and resealing can be done easily while the shear is mounted to the machine.
- Service and adjust the pivot group without the need for any special tools.
- Easily change the tip and flip the blades in thirty minutes with standard tools.
- Access the cylinder's hoses and speed valve easily from the side of the shear.
- You are supported. The Cat Dealer Network is your one point of contact for all your service needs.

Options to Meet Your Needs

- Connecting your shear has never been easier. Dual high-pressure ports attach to any compatible machine regardless of the hydraulic layout.
- Meet your specific needs with straight and rotating configurations and boom and stick mount options.

  Straight – S2000 Series
  - – Boom mounted
  - – Used in secondary demolition and scrap applications
  - – Rotator is eliminated, reducing maintenance costs
  - – Larger shears can be used on smaller machine size classes

  Rotating – S3000 Series
  - – Boom or Stick mounted
  - – Used in scrap, primary, and secondary demolition applications
  - – Provides optimal cycle times and positioning
- Boom Mounted – Ideal for scrap processing, allows use of a larger shear for increased production.
- Stick Mounted – Ideal for demolition applications, maximizes reach for structural demolition.

Machines (HEX)	304	305	307	308	311	312	313	314	315	316	318	320	323	325	326	329-330	336-340	349-352	374	385-390
Model																				
S305	B	B	S	S																
S3015			B	B	S	S	S	S	S	S	S									
S3025										B	B	S	S							
S3035												B	B	B	S	S	S			
S2050 Straight												B	B							
S3050															B	B				
S2070 Straight																B	B			
S3070																	B	B	S	S
S2090 Straight																	B	B		
S3090																		B	S	S

Primary Match | **B** Boom Mount | **S** Stick Mount

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